Sam Shelly

Weed, CA | (530) 340-1065 | samgshelly@gmail.com | linkedin.com/in/sam-g-shelly | github.com/SamZombie | samzombie.github.io

Education

Vassar College — Aug 2022 – May 2026
Bachelor of Arts in Computer Science, Minor in Astronomy — Poughkeepsie, NY

GPA: 3.64/4.00 | Major GPA: 3.68/4.00

Relevant Coursework: Natural Language Processing, Numerical Methods, Analysis of Algorithms, Compilers, Theory of Computation, Data Structures & Algorithms

Experience

AI Development Intern — May 2025 – Jan 2026
Innovative Computing Systems — Los Angeles, CA

- Led a four person team to architect a Microsoft Copilot Studio knowledge agent for a 60-person systems integration firm.
- Restricted the agent to closed, company-fed data sources to protect client IP and eliminate public internet exposure.
- Built Python ingestion pipelines and ConnectWise API integrations to power AI-driven enterprise search, reducing information retrieval time by 30%.
- Launched a 15-user pilot and led technical evaluations for clients, advancing the product towards commercialization.

Computer Science Department Coach — Jan 2024 – May 2026
Vassar College — Poughkeepsie, NY

- Led three sessions per week on algorithms, data structures, and debugging in Java for up to 50 students per semester (200+ total), adapting pacing and session plans for struggling students.
- Mentored two coaches, and met weekly with the course professor to align on curriculum direction and student learning progress.

Projects

AI Knowledge Base RAG — GitHub
Python, FastAPI, LangChain, Qdrant, Ollama, Mistral 7B, BAAI/bge-large-en-v1.5, PyMuPDF, Docker, Streamlit, pytest, GitHub Actions

- Built a closed-loop Retrieval-Augmented Generation (RAG) system using self-hosted, open-weight components with no external API calls to securely query government and defense AI policy documents in air-gapped and other privacy-sensitive environments.
- Developed a custom LLM-as-judge evaluation pipeline achieving 0.95 retrieval recall and 0.92 answer similarity across a ground-truth QA dataset, with results visualized in a Streamlit dashboard.
- Built an end-to-end document ingestion and retrieval pipeline using PyMuPDF, semantic chunking, BAAI/bge-large-en-v1.5 embeddings, Qdrant vector search, and Mistral 7B (Ollama) for grounded, cited responses.
- Deployed the application as a Dockerized FastAPI REST service with automated testing and CI/CD using GitHub Actions.

Publications

Feedback-Driven Magnetic Field Evolution in MW-type Simulated Circumgalactic Medium — 2024
Param P. Gogia, Umman Azan, **Sam Shelly**, Edward Buie II | KNAC Undergraduate Symposium Proceedings

- Investigated turbulence-driven feedback's role in magnetic field evolution in the CGM of a simulated Milky Way-type galaxy across three MAIHEM MHD simulations (two solar-metallicity, one at 0.3 $Z_{\odot}$).
- Found that feedback induces late-stage magnetic field saturation via disk instability and outward propagation of cold, magnetized gas, versus steady cubic growth in the no-feedback case.
- Analyzed post-simulation Mach-Alfvén profiles, angular momentum phase plots, magnetic/kinetic energy density evolution, and built data visualizations.

Efficacy of NER Models on Fiction and Nonfiction Domains — 2025
Duncan Beauchamp, **Sam Shelly** | CMPU 366 Natural Language Processing, Vassar College

- Evaluated cross-domain performance of a DistilBERT-based NER model on literary fiction (LitBank) vs. expository text (OntoNotes), restricted to shared PER/LOC/ORG entity types.
- Reduced F1 domain gap between genres from 0.368 to 0.128 by fine-tuning on a mixed-domain, raising LitBank micro-F1 from 0.26 to 0.72.
- Built the training data pipeline which involves dataset loading, chunking, oversampling, and train/validation/test splits.

Skills

Languages: Python, Java, C, C++, OCaml | **ML/AI & Data:** PyTorch, HuggingFace Transformers, DistilBERT, scikit-learn, LangChain, Ollama, Qdrant, pandas, yt-project | **Tools:** Docker, FastAPI, GitHub Actions, PyMuPDF